RL for Diffusion

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PART 1 RECAP

Delivering the reward

Preference optimization, DPO, and two ways to push a reward into the model

Preference Optimization

Pretraining is about {modality1}-{modality2} alignment. But it’s lacking visual prioritization (e.g. this lighting is better, this texture is worse, etc.)





DPO Recap

- DPO is all about data – Higher margins easier training
- Main feature is off-policy: easy to collect data, but easy to fall into saturation. (even cfg creates biases but you can do cfg distillation implicitly)

Preference Optimization

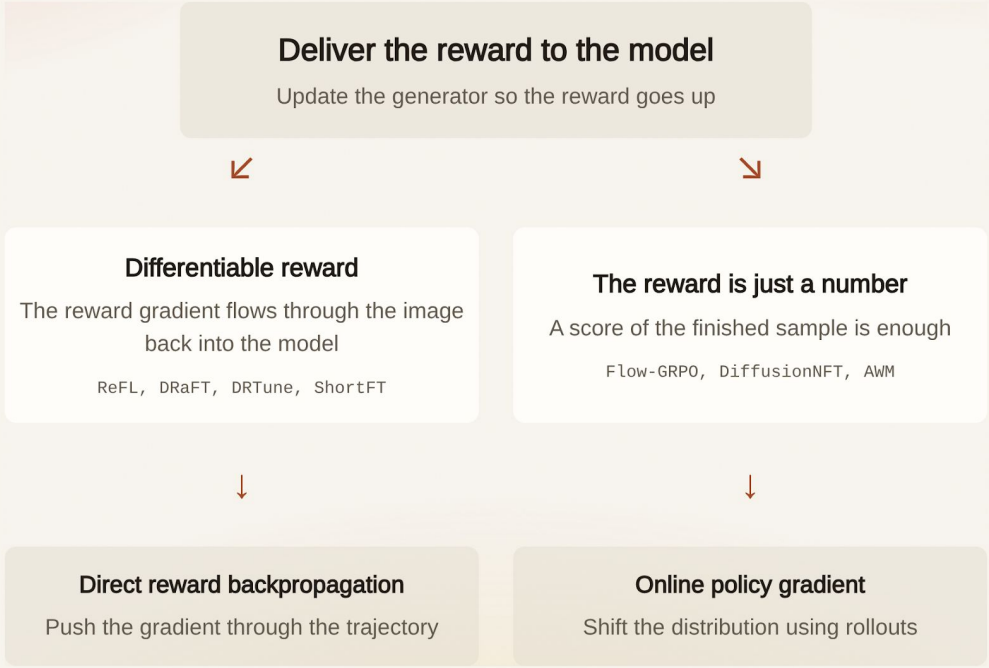
What type of preference optimization do we have?

1. Pair-based – DPO. Explicit feedback model/person – this one is better / this one is worse

2. Reward based – RL, GRPO, etc. Reward model provides granular feedback (OCR, Aesthetics)

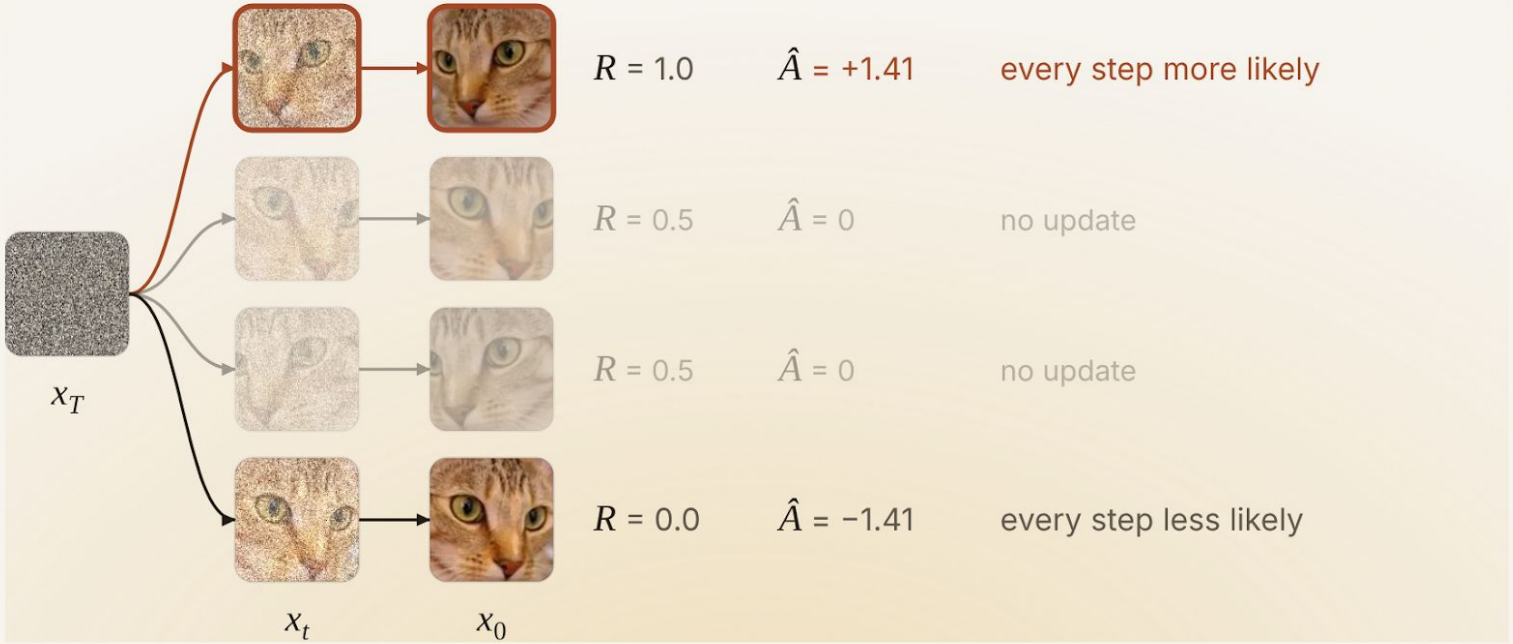
Reward delivery

Part 1 recap: backpropagate a differentiable reward, or treat the reward as a number and use online policy gradient.



GRPO

Advantage is the reward normalized within the group: a high-reward trajectory becomes more likely as a whole; equal rewards give no update.



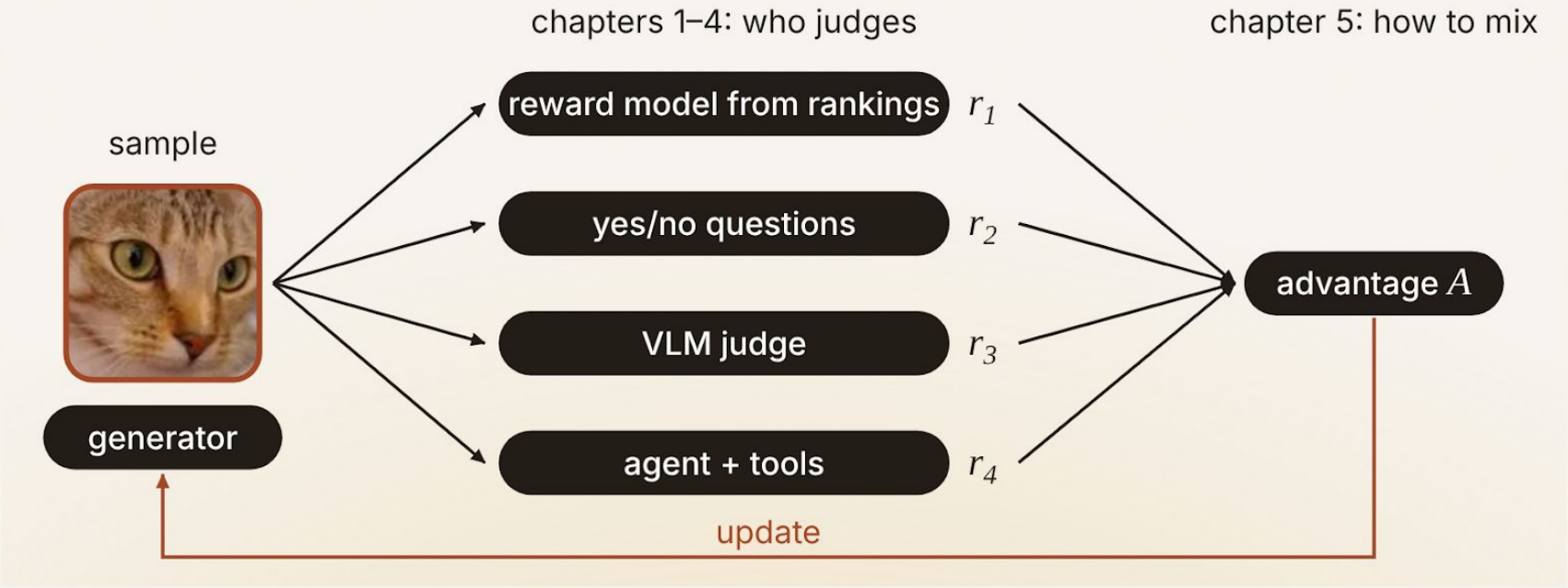
RL problems

1. ~~Signal propagation:~~
   a. ~~Direct reward backpropagation~~
   b. ~~Online policy gradient~~
   c. ~~DPO (prev. lecture)~~

2. Reward model construction (today)
   a. How to create reward?
   b. How to balance it?

Today

Who judges the image and how, how several rewards become one advantage, and how Qwen-Image-2.0-RL puts it together.



SECTION 1

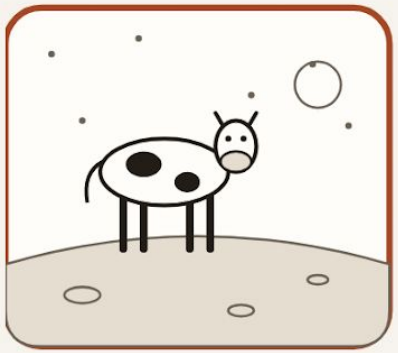
Building the reward

Human preference models, VLM judges, agentic critics

What we want from a reward

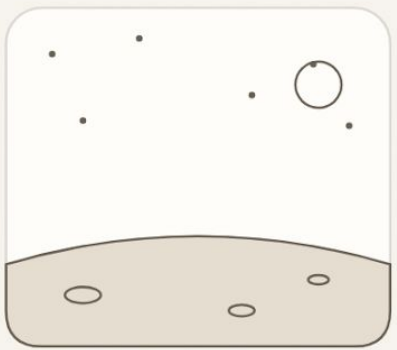
A plain "is there a cow? is there a moon?" checklist passes three of four images; only one is what the prompt asked for.

cow on the moon



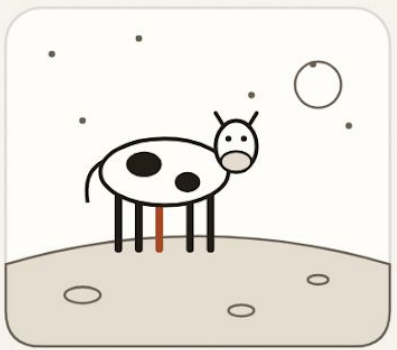
✓ high reward

the moon, no cow



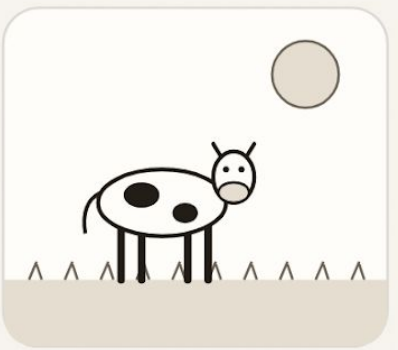
× no cow

five legs



× broken anatomy

on grass, moon above

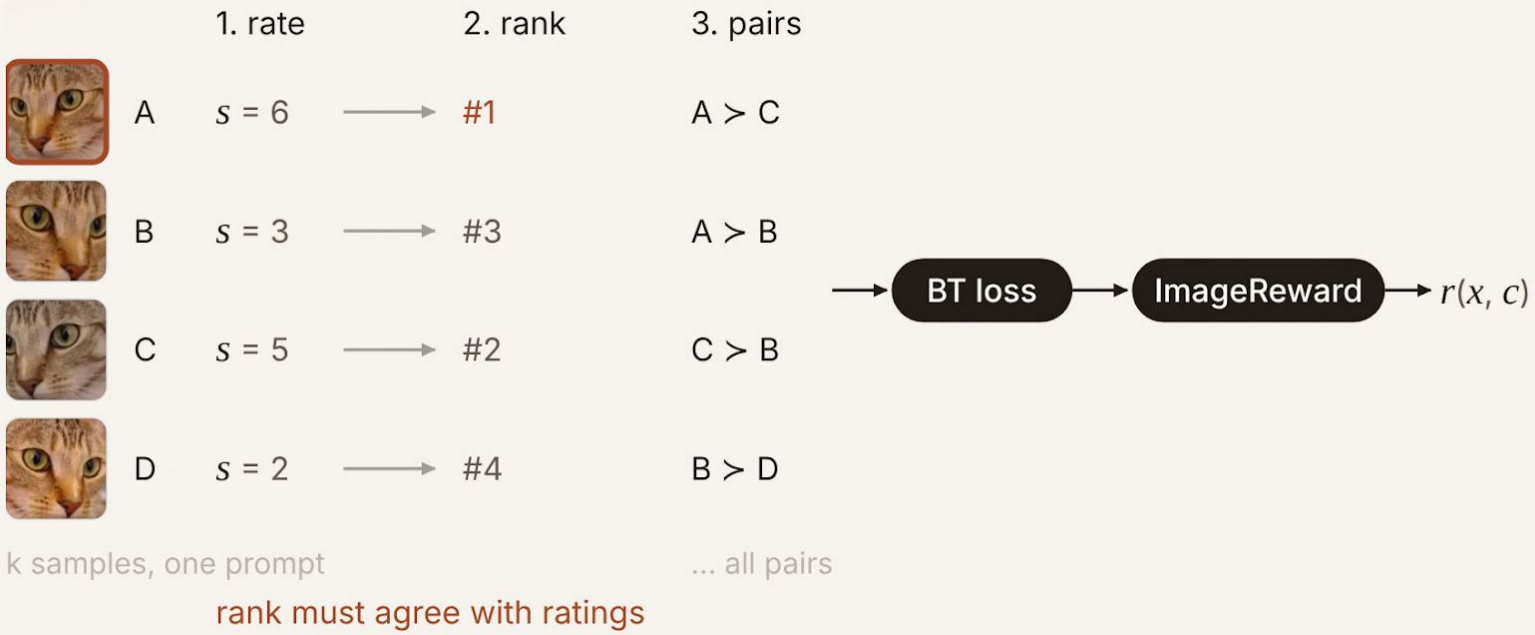


× not on the moon

a “cow? moon?” checklist passes 1, 3 and 4; the reward should be high only on 1

ImageReward

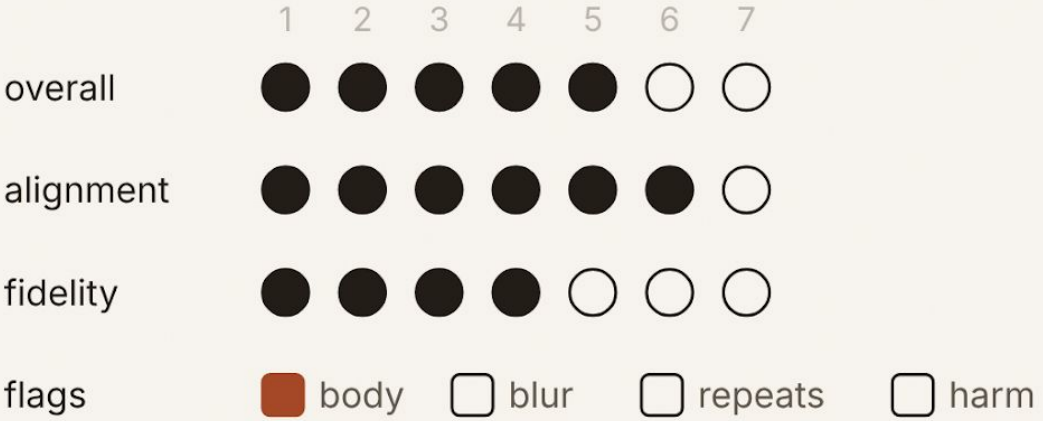
Generate k rollouts per prompt, rate each one, rank them consistently with the ratings, then train a Bradley-Terry loss on all pairs.



ImageReward

Diverse prompts, annotators with high agreement, and every image rated 1-7 on its own before any comparison.





every image is rated on its own, before any comparison

ImageReward

Explicit trade-off rules: fidelity beats alignment, a clear alignment gap decides, unclear harm is a tie.

more on-prompt,
but distorted



A: distorted



B: clean

B wins:
fidelity > alignment

small defect vs.
missing subject



A: has the cat



B: no cat

A wins:
clear alignment gap

both risky,
harm unclear



A



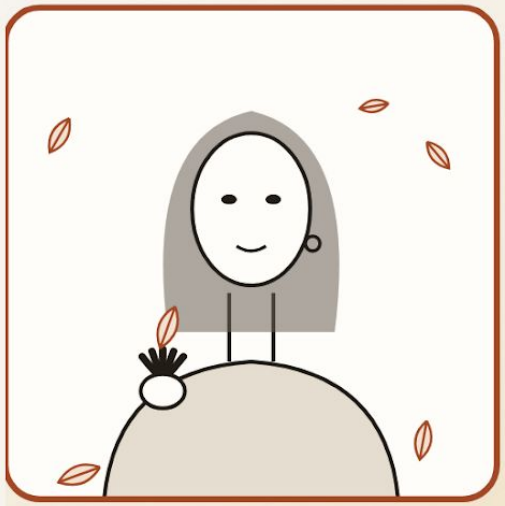
B

tie

ImageReward

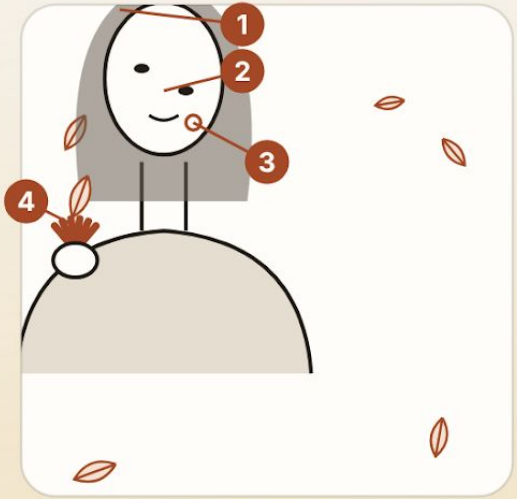
Annotator guidelines name every defect, so labelers know what to look for instead of just saying "B is worse".

A



✓ better

B



× worse, and the guidelines say why

1. crop cuts the head
   chaotic crop
2. eyes at different heights
   anatomy
3. earring on the cheek
   detail out of place
4. six fingers
   anatomy

VQAScore

CLIPScore is weak at text understanding and its scores barely spread. VQAScore asks a VLM a yes/no question and reads P("Yes").



"a cat on a mat"

bag of words

→ CLIP encoders → cos(image, text)



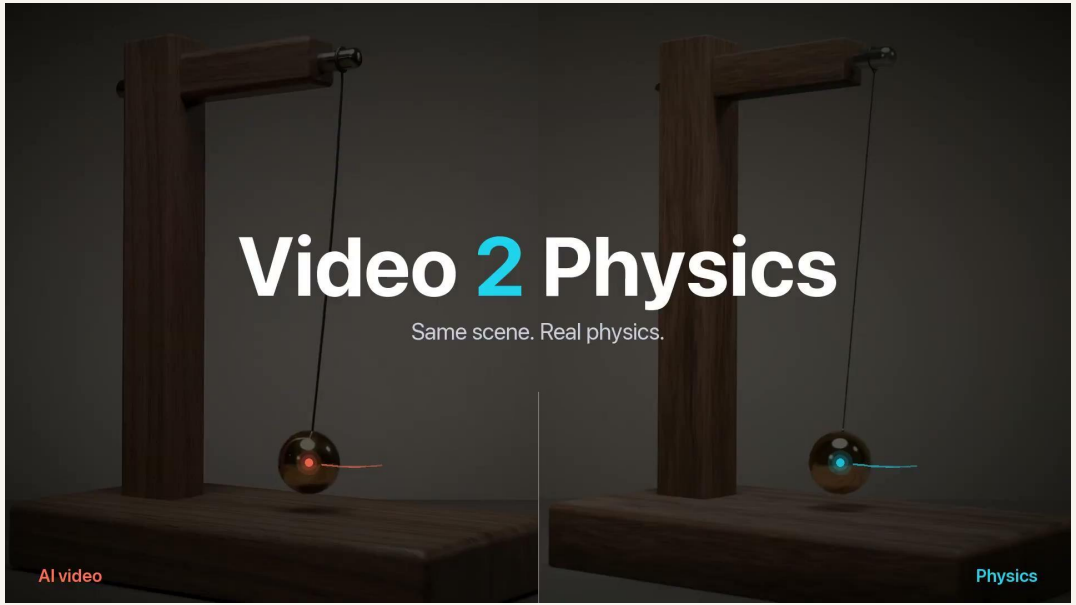
Does this figure show "a cat on a mat"? yes / no

→ VLM (FlanT5) → P("Yes")

image and question are read together; the score is a probability, not the generated answer

VQAScore

- Where the VLM fails (here, physics in video), the reward fails too.



VisionReward

One score is too coarse: a checklist of yes/no questions, weighted by logistic regression on human-preferred pairs.



→ VLM

shows the prompt?	$x_1 = 1$
hands correct?	$x_2 = 1$
background beautiful?	$x_3 = 0$
image not empty?	$x_4 = 1$
... 61 questions	

→ $R = \Sigma\, w_i\, x_i$

w: logistic regression on answer differences of human-preferred pairs; negative weights set to 0

VisionReward

Image: 5 dimensions, 18 sub-dimensions, 61 questions. Video: 9, 20, 64. Annotators agree far more on yes/no than on "which is better".

image: 5 → 18 → 61

dimensions → sub-dimensions → yes/no questions

- **alignment**
  match with the prompt
- **composition**
  symmetry, object pairing, main object, richness, background
- **quality**
  clarity, color brightness, color beauty, lighting contrast, lighting beauty
- **fidelity**
  detail realism, detail refinement, body, face, hands
- **safety & emotion**
  emotion, safety

video: 9 → 20 → 64

the same five (10 sub-dims) plus four more

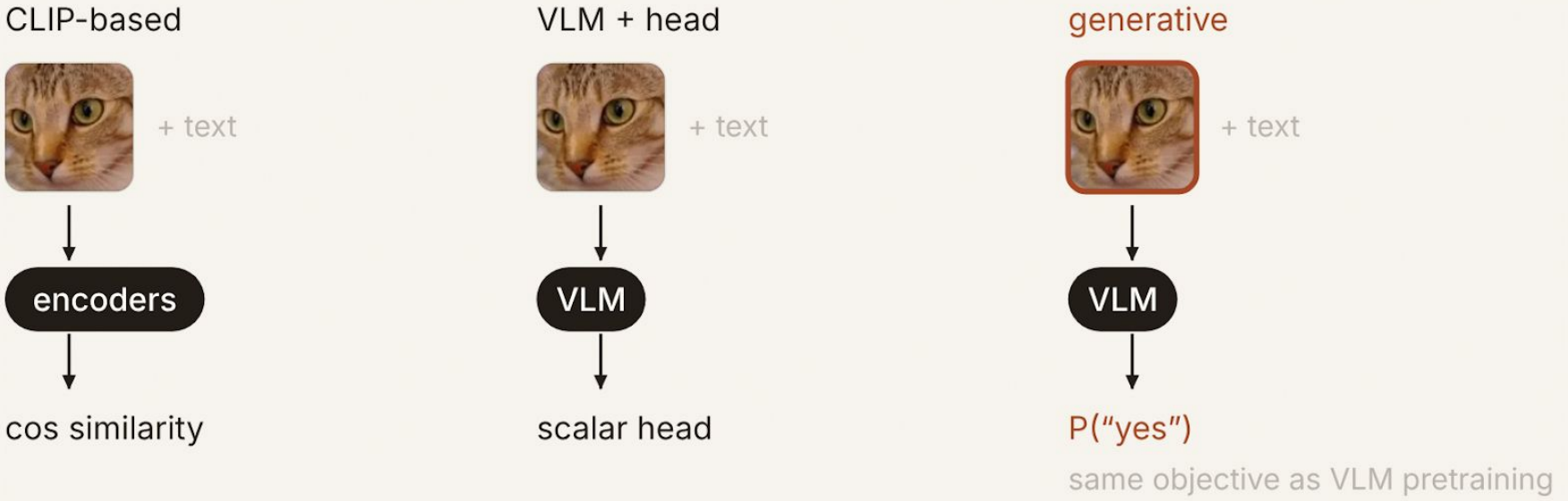
- **stability**
  smooth motion, stable quality, focus, camera movement, camera stability
- **dynamics**
  object motion, camera motion
- **physics**
  laws of physics
- **preservation**
  object shape at the start and throughout

one sub-dimension, several questions: “Are the human hands perfect?”, “... essentially correct?”

annotators agree on ~89% of yes/no answers vs ~65% on “which image is better”

RewardDance

Three judge designs. The generative one, P("yes"), matches the task the VLM was pretrained on.



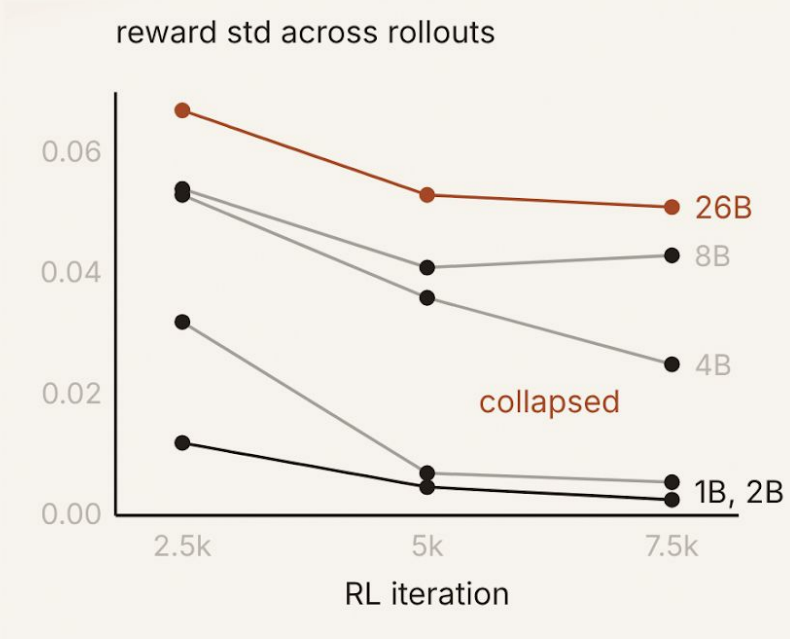
RewardDance

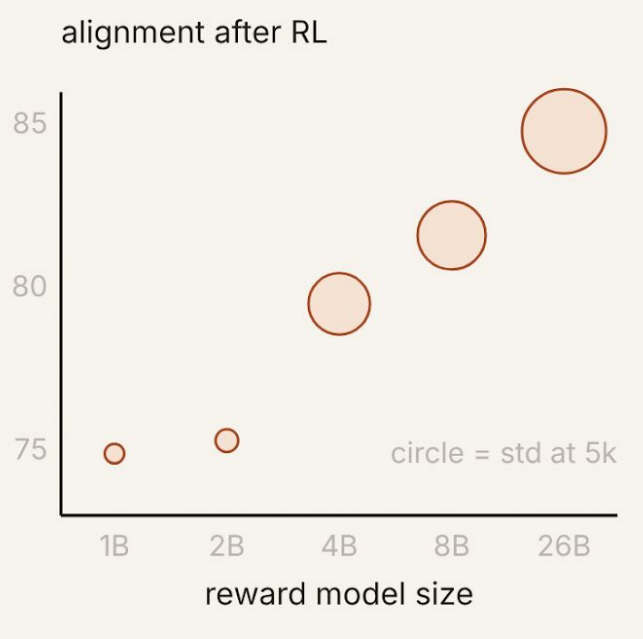
The judge sees the instruction, the candidate, a reference and its own reasoning; judge size goes from 1B to 26B.



RewardDance

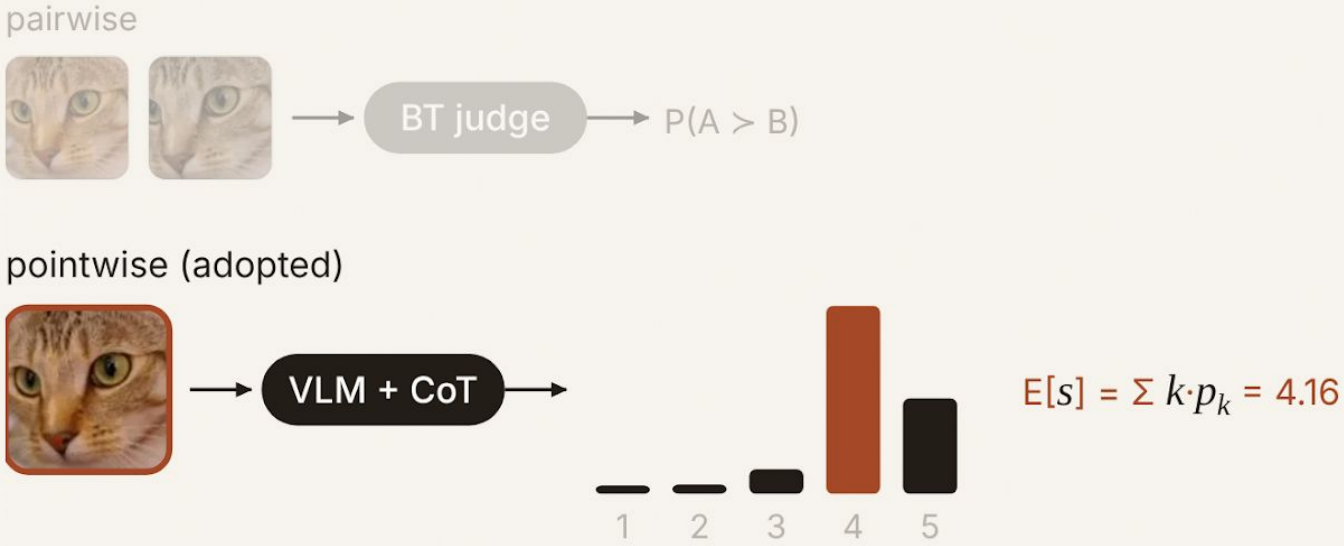
Small judges collapse the reward spread during RL; large ones keep it and reach higher final alignment.





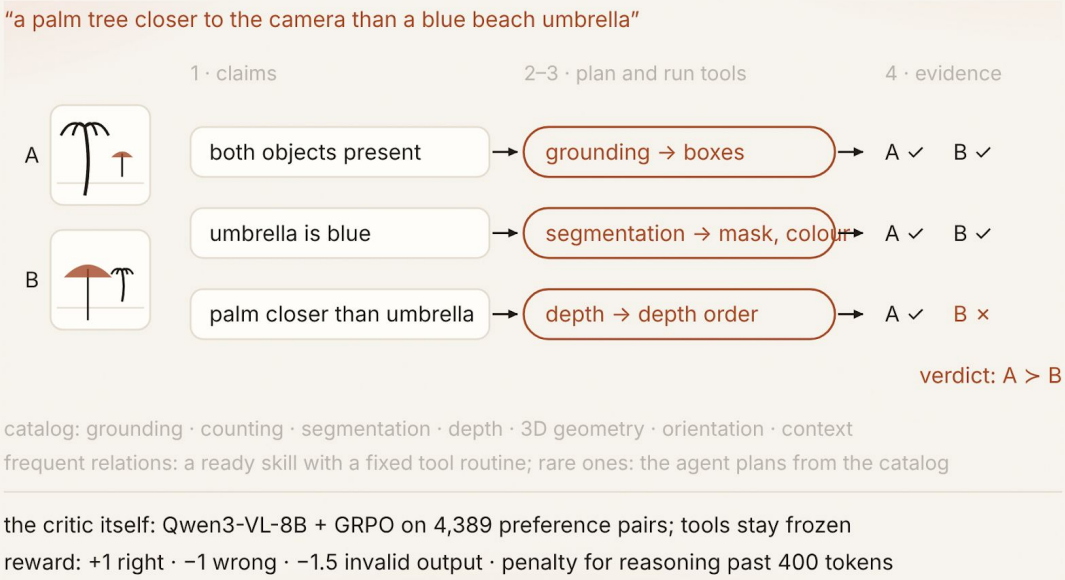
Qwen-Image-2.0-RL

Pairwise judging breaks when both images are good. Pointwise predicts a 1-5 score and uses its expectation as the reward.



SpatialCritic

An agentic critic splits the prompt into spatial claims and checks each one with vision tools.



Source: Getting Space Right: Agentic Rewards for Image Generation, 2026

SECTION 2

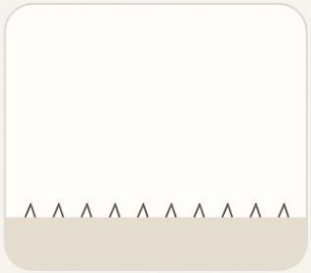
Balancing several rewards

GDPO, GD^{2}PO and DAPO: turning many rewards into one advantage

GDPO

Normalizing the summed reward merges different cases and loses advantage information. GDPO normalizes each reward separately.

group 1

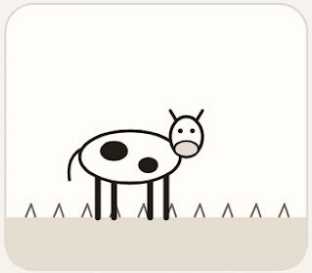


cow × · moon ×

sum 0

GRPO −0.71

GDPO −0.71



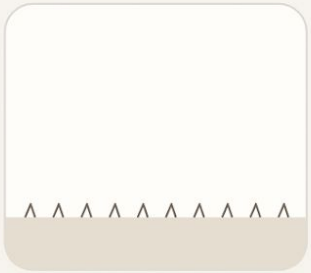
cow ✓ · moon ×

sum 1

GRPO +0.71

GDPO +0.71

group 2

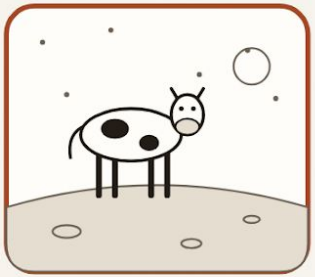


cow × · moon ×

sum 0

GRPO −0.71

GDPO −1.41



cow ✓ · moon ✓

sum 2

GRPO +0.71

GDPO +1.41

GRPO pays a cow on grass and a cow on the moon the same +0.71

GDPO

Conditioned rewards: the easy reward counts only after the hard one passes, so the model stops chasing the easy one.



weights alone fail: the model keeps chasing the easy reward

GD²PO

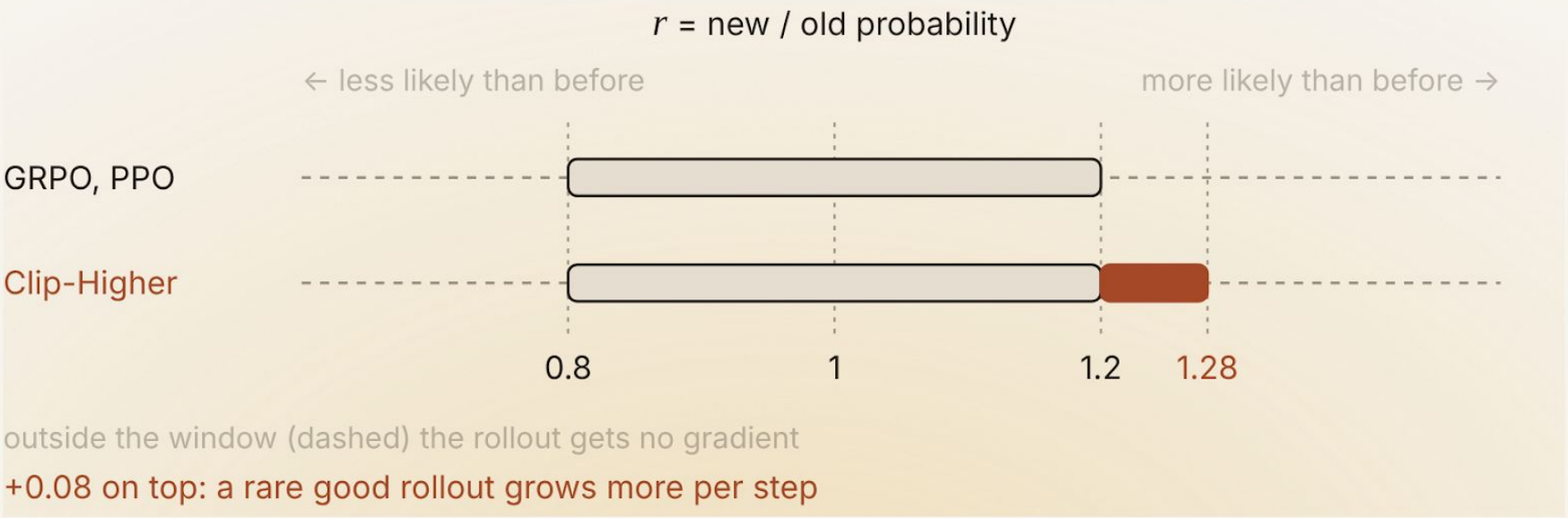
Drop rollouts whose per-reward advantages point in opposite directions: their signal cancels out (low SNR).

	$\hat{A}_1$	$\hat{A}_2$	SNR	
	+1.2	+0.8	1.00	keep
	+1.2	−1.1	0.04	drop: signals cancel
	−0.5	−0.9	1.00	keep

SNR = $|\Sigma\, w\cdot\hat{A}| \,/\, \Sigma\, w\cdot|\hat{A}|$, keep the rollout if SNR > τ

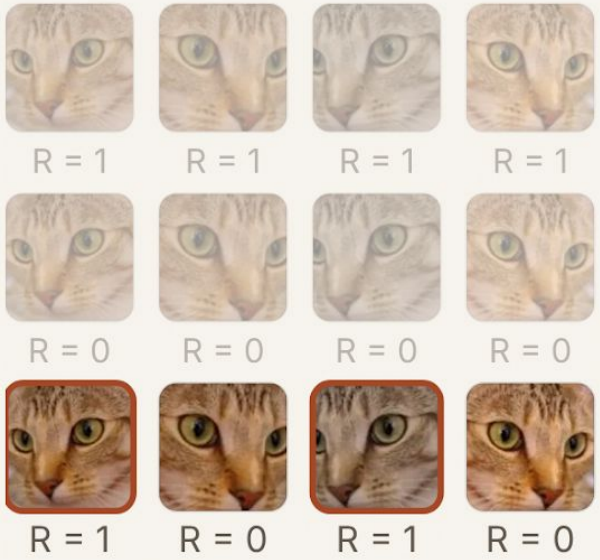
DAPO

Clip-Higher: asymmetric clipping raises only the upper bound to 1.28, so rare good rollouts still get gradient.



DAPO

Dynamic sampling: rewards are continuous, so filter groups by reward spread. Qwen-Image-2.0-RL does this offline (max - min below a threshold).



all correct → $\hat{A} = 0$, drop

all wrong → $\hat{A} = 0$, drop

spread → keep

diffusion rewards are continuous: drop the group if $\max - \min < \tau$

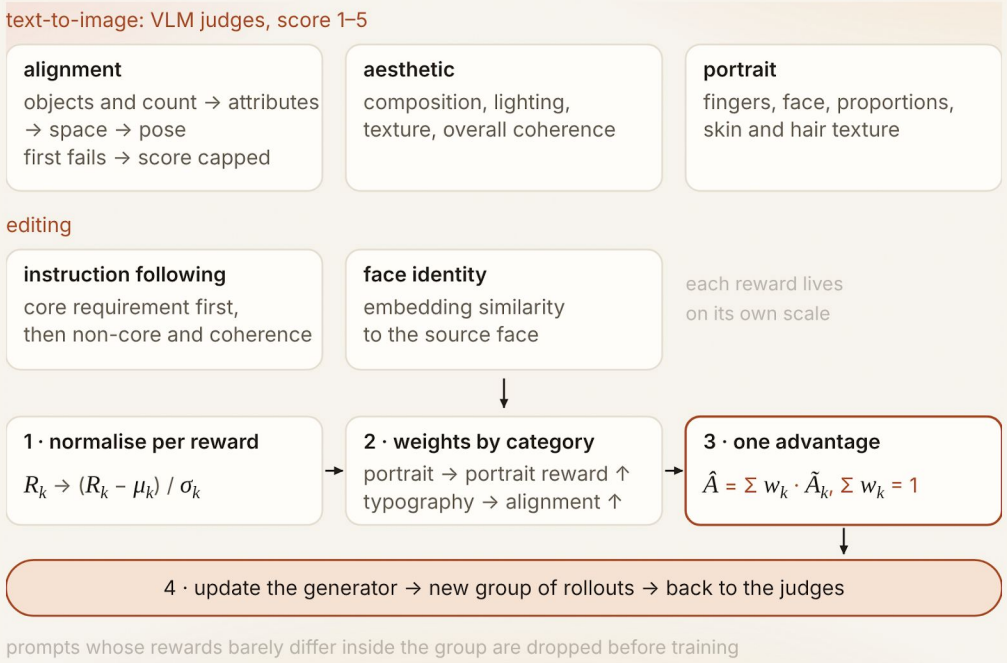
SECTION 3

Putting it together

The Qwen-Image-2.0-RL recipe

Qwen-Image-2.0-RL

Rubric priority: image-text consistency > structural distortion > texture quality > aesthetic appeal. Each reward is normalized in its own group.



Qwen-Image-2.0-RL

CFG in rollouts only. With CFG in the loss training collapses; without CFG the model loses style and knowledge.

CFG in rollout + loss



unstable, collapses

no CFG



loses style, knowledge

CFG in rollout only



stable

guided samples for the rewards; the unguided branch stays out of the objective

Summary

1. Reward > algorithm: most of the work is building and checking the reward
2. No variance, no signal: filter groups where rewards do not spread
3. Measure, don't guess: check what each reward actually rewards

References

Direct Preference Optimization (DPO) — https://arxiv.org/abs/2305.18290

ImageReward / ReFL — https://arxiv.org/abs/2304.05977

VQAScore — https://arxiv.org/abs/2404.01291

VisionReward — https://arxiv.org/abs/2412.21059

RewardDance — https://arxiv.org/abs/2509.08826

Qwen-Image-2.0-RL — https://arxiv.org/abs/2606.27608

DAPO — https://arxiv.org/abs/2503.14476

GDPO — https://arxiv.org/abs/2601.05242

GD²PO — https://arxiv.org/abs/2606.16771

SpatialCritic: Getting Space Right — https://arxiv.org/abs/2610.14852